

Statistical learning theory

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

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Section 1

Empirical risk minimisation

Vladimir Vapnik



Vladimir Vapnik



Second edition, 2006

- He showed that catching spam, fitting a price and estimating a probability are **one problem**, scored three ways.
- **Catching spam.** The score counts how often the answer is wrong.
- **Fitting a price.** The score is the squared miss, the one this room has used all day.
- **Estimating a probability.** The score is how surprised the model is by what actually happened.
- Three problems that had lived in different books. The score says which one you are doing, and what happens to the score afterwards never changes.

Estimation of Dependences Based on Empirical Data. Russian original 1979, Kotz translation Springer 1982, second edition 2006.

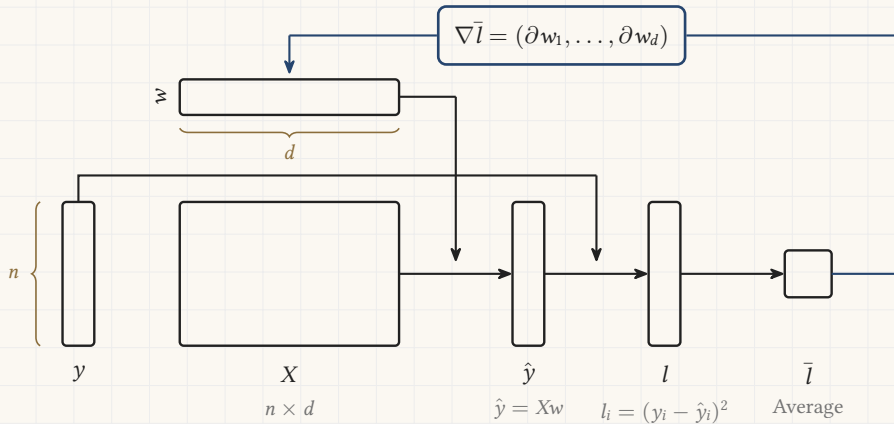
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Section 2

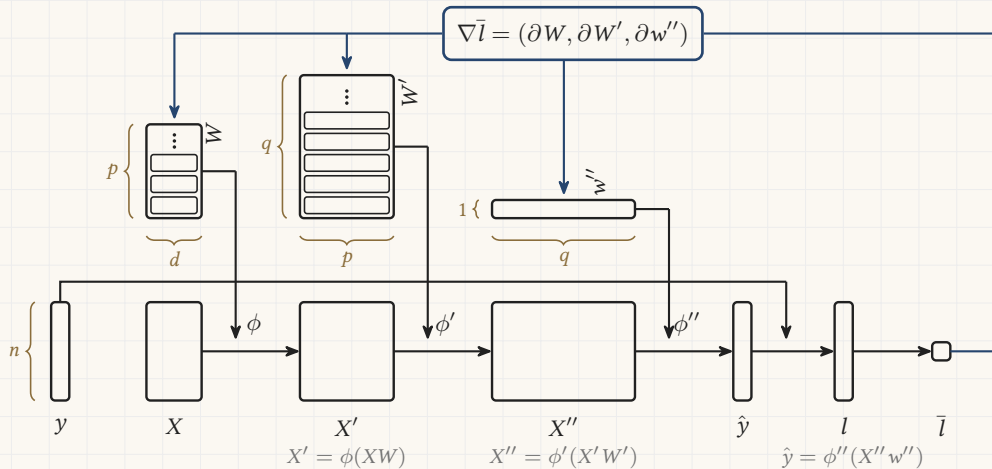
The loop we have been running

Linear regression with gradient descent

Two o'clock, in one picture. Each box is a piece of data, and its shape is the shape of the data.

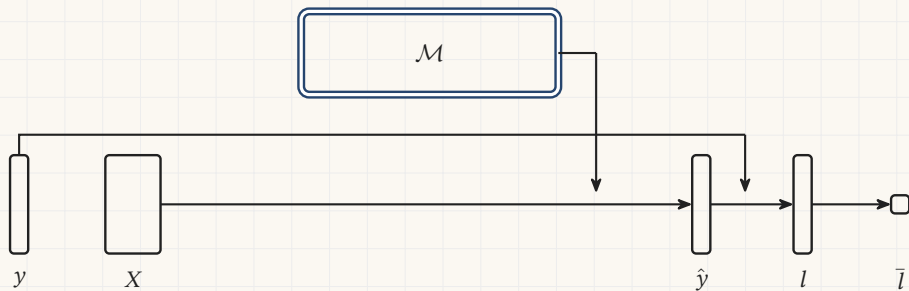


A neural network



Everything between the data and the estimate

The three weight blocks and the two tables between them are the model. Give the whole of it one name.



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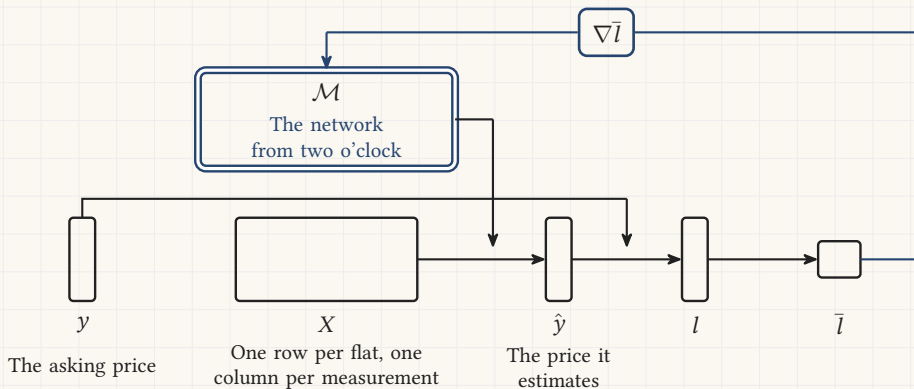
Section 3

The same loop, elsewhere

The loop, with the model as one box

The same picture, with $\mathcal{M}$ where the network was.

The five flats are the ones from this morning, and the network is the one from this afternoon.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.

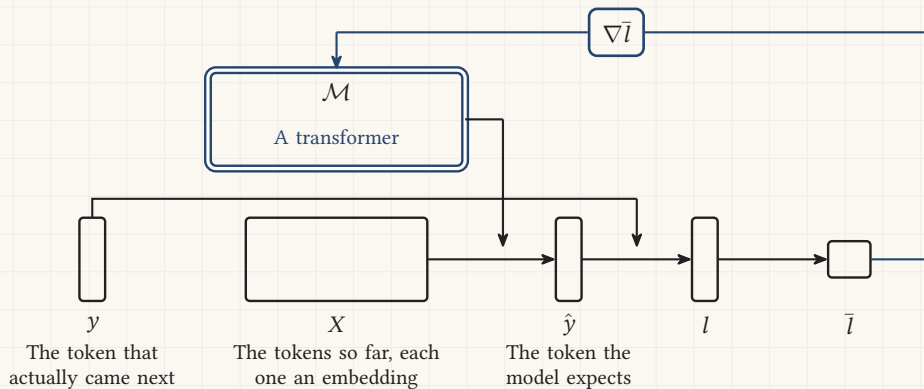
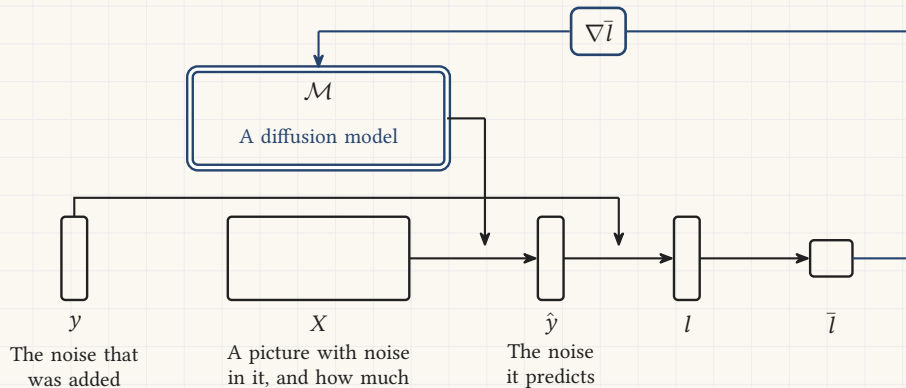


Image generation

Add a measured amount of noise to a picture, then ask which noise was added.

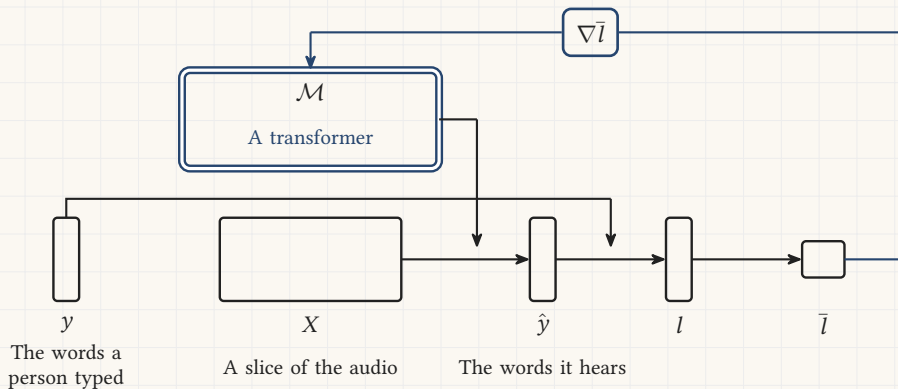
The loss here really is the squared miss. Ho, Jain and Abbeel, 2020; Rombach and others, 2022.



Transcription

Every recording that already carries a transcript is a row. There are 680 000 hours.

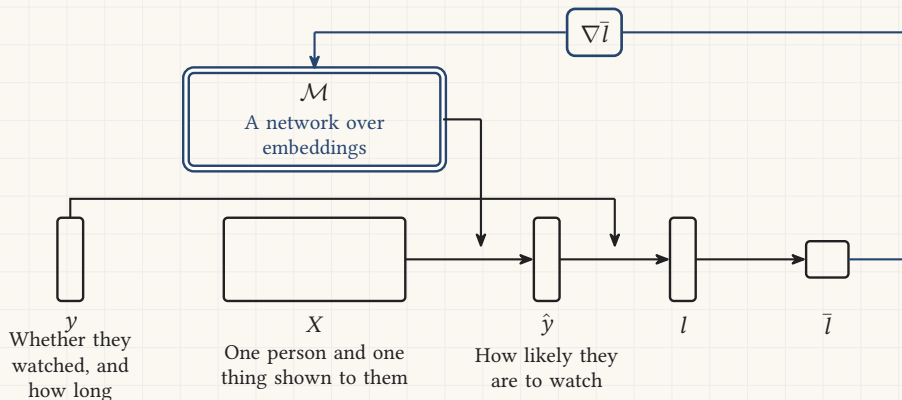
Whisper. Radford and others, ICML 2023.



Ranking and recommendation

Every impression is a row: what was shown, to whom, and whether they took it.

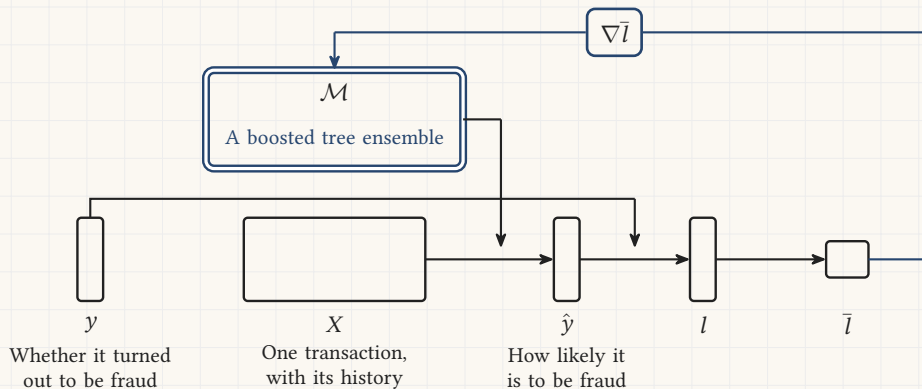
The system that picks the next video. Covington, Adams and Sargin, RecSys 2016.



Fraud and credit decisions

Every settled transaction is a row, and its label arrives weeks later with the chargeback.

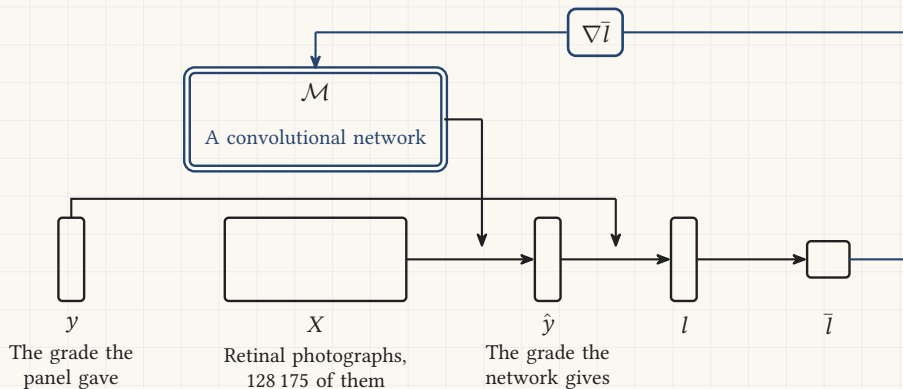
Dal Pozzolo and others, 2018, with Worldline. The tree ensemble is fitted to the gradient too.



Screening for diabetic eye disease

In 2018 a regulator cleared a machine to read these photographs with no doctor behind it.

128 175 photographs, graded 3 to 7 times by 54 ophthalmologists. Gulshan and others, *JAMA*, 2016.



Four outcomes

The trial behind that clearance: 819 people at ten primary care clinics.

Abràmoff and others, *npj Digital Medicine*, 2018. Counts from its figure 1.

		<i>What the experts found</i>	
		Has the disease 198 people	Does not 621 people
<i>What the machine said</i>	Refer	173 Caught True positive	65 False alarm False positive
	All clear	25 Missed False negative	556 Rightly cleared True negative

Precision and recall

Two questions to ask of any machine that answers yes or no.

Medicine calls recall sensitivity. Abràmoff and others, *npj Digital Medicine*, 2018.

What the experts found			
		Has the disease 198 people	Does not 621 people
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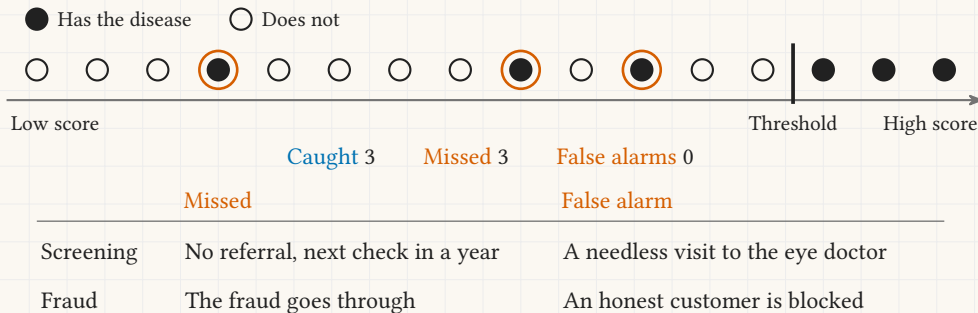
Precision: of the 238 it referred, 173 had it, about 73 in 100

Recall: of the 198 who had it, it caught 173, about 87 in 100

The threshold

The machine gives every case a score. People choose the score from which it says yes.

A schematic. In the trial the threshold was fixed before the first patient.



References: Vapnik and the language models

Vladimir N. Vapnik. *Estimation of Dependences Based on Empirical Data*. Translated by Samuel Kotz. Springer, 1982; second edition 2006.

doi.org/10.1007/0-387-34239-7

Ashish Vaswani and others. Attention is all you need. *Advances in Neural Information Processing Systems* 30, 2017.

arxiv.org/abs/1706.03762

Long Ouyang and others. Training language models to follow instructions with human feedback, 2022.

arxiv.org/abs/2203.02155

DeepSeek-AI. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* 645, 633 to 638, 2025.

arxiv.org/abs/2501.12948

References: the other applications

Jonathan Ho, Ajay Jain and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS* 33, 2020.

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Robin Rombach and others. High-resolution image synthesis with latent diffusion models. *CVPR*, 2022.

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Alec Radford and others. Robust speech recognition via large-scale weak supervision. *ICML*, PMLR 202, 28492, 2023.

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Paul Covington, Jay Adams and Emre Sargin. Deep neural networks for YouTube recommendations. *RecSys*, 191 to 198, 2016. doi.org/10.1145/2959100.2959190

Andrea Dal Pozzolo and others. Credit card fraud detection: a realistic modeling and a novel learning strategy. *IEEE Transactions on Neural Networks and Learning Systems* 29(8), 3784 to 3797, 2018.

doi.org/10.1109/TNNLS.2017.2736643

Tianqi Chen and Carlos Guestrin. XGBoost: a scalable tree boosting system. *KDD*, 2016. arxiv.org/abs/1603.02754

Varun Gulshan and others. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA* 316(22), 2402 to 2410, 2016. doi.org/10.1001/jama.2016.17216

References: the screening trial

Michael D. Abràmoff and others. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *npj Digital Medicine* 1, 39, 2018. doi.org/10.1038/s41746-018-0040-6

From LLMs to reasoning models

Agentic AI, part 1

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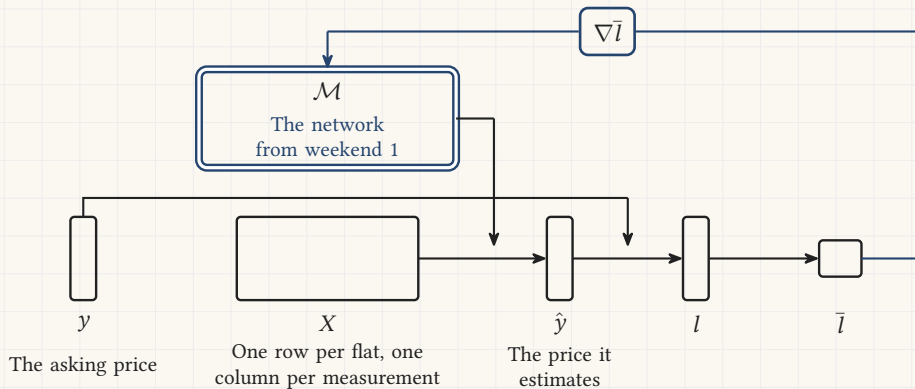
Section 1

How a language model is trained

The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

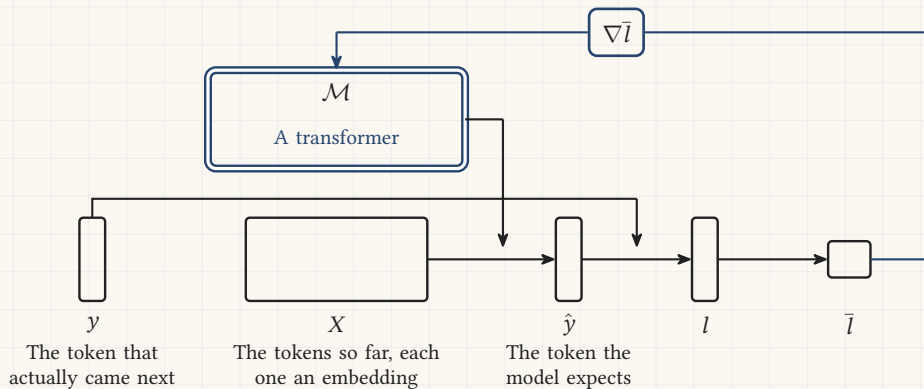
The five flats and the network are the ones from weekend 1.



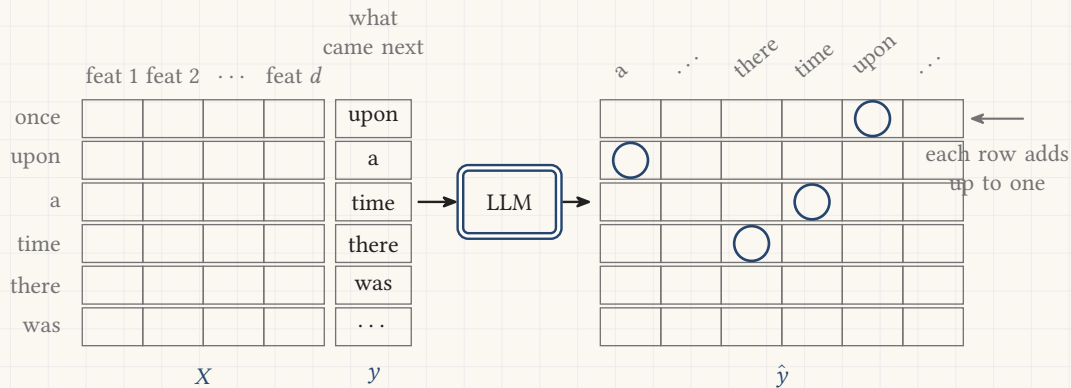
Large language models

A row is text with its next token cut off. Every position of every document gives one.

The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



One row for every position



Reading that matrix

Every row One position of the context. The sentence has six words, so $\hat{y}$ has six rows.

Every column One token of the vocabulary. Every word in the dictionary gets a column, whether or not it appears in the sentence.

Every entry $\hat{y}_{ij}$ is how strongly the model believes the word at position i should be word j .

$$\sum_j \hat{y}_{ij} = 1 \quad \text{for every row } i$$

So the output is a whole table of beliefs, one for every position at once. Predicting the next word is reading a single row of it.

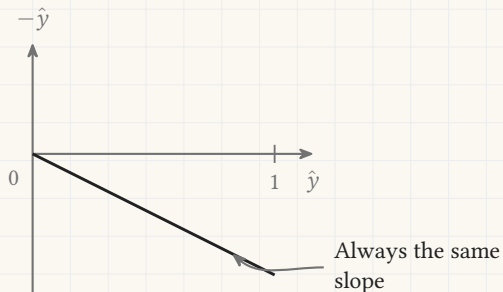
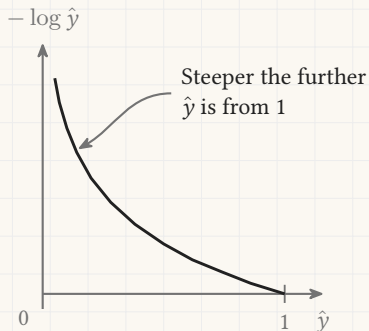
Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

$$\ell = -\log \hat{y}_{1, \text{"upon"}} - \log \hat{y}_{2, \text{"a"}} - \log \hat{y}_{3, \text{"time"}} - \dots$$

Both versions are smallest when the model puts all of its weight on the word that actually came next.
They differ in how hard they push when it does not.

Why the logarithm



The slope is the gradient. A constant one pushes just as gently on a word the model gave no chance as on one it nearly had right.

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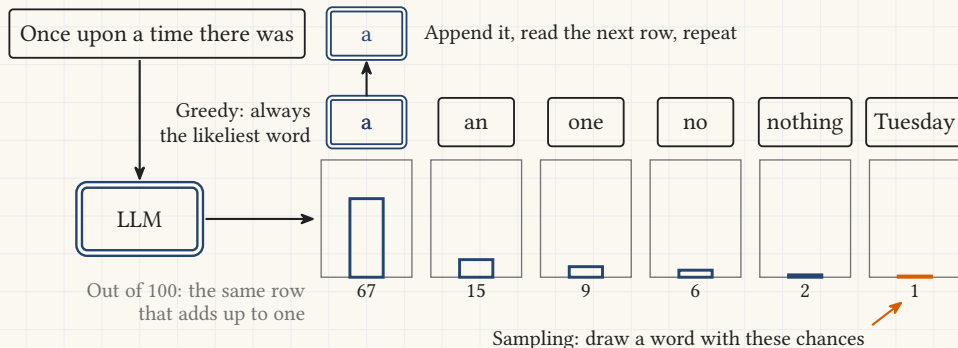
Section 2

How a language model writes text

Greedy decoding and sampling

To write, the model reads one row: the row after the text so far.

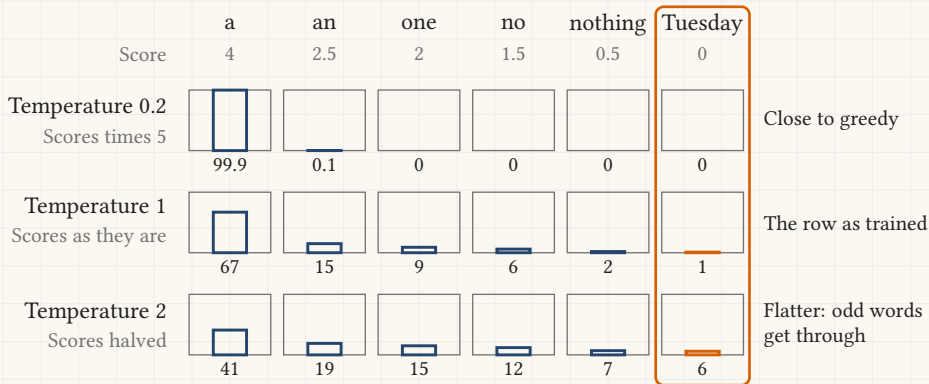
Six words stand in for the dictionary. The numbers are made up.



Temperature

Behind the row is a score for every word. Every score is divided by the temperature first.

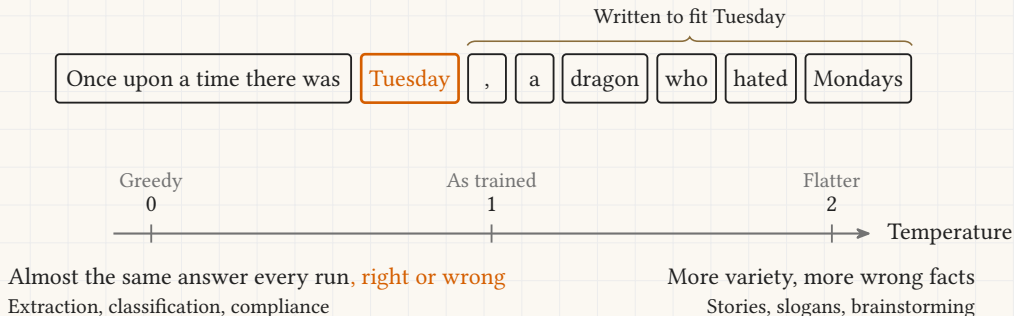
Jurafsky and Martin, *Speech and Language Processing*, 3rd edition draft, 2026, section 7.6.



Temperature in practice

A word, once written, stays. Every later row is read after it.

Holtzman and others, ICLR 2020. Zhang and others, *How language model hallucinations can snowball*, ICML 2024.



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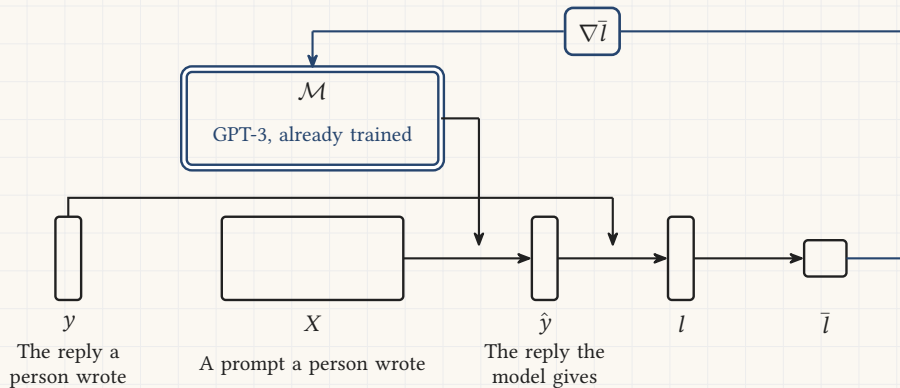
Section 3

How a language model becomes a chatbot

Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



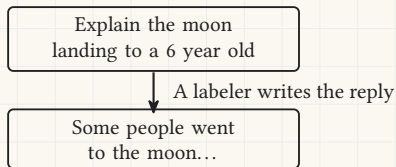
The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

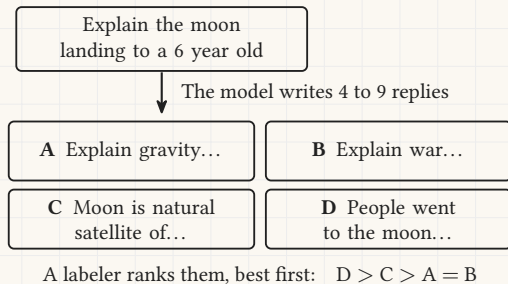
One demonstration

12 725 prompts, most of them written by the labelers



One comparison

33 207 prompts, most of them sent by users of the API



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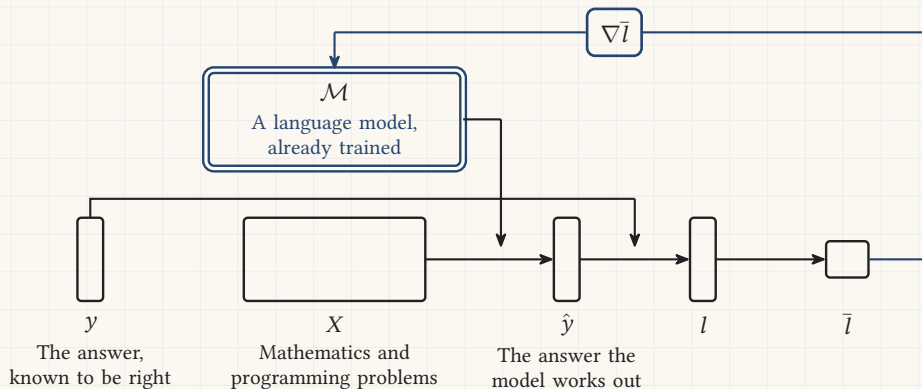
Section 4

How a chatbot becomes a reasoning model

Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



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Section 5

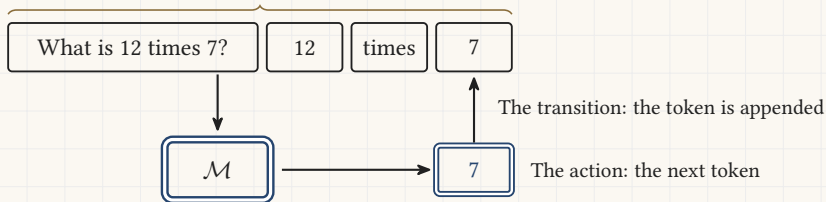
Reinforcement learning

A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

The state: the prompt and the tokens written so far



The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward	
What is 12 times 7?	12					0	
What is 12 times 7?	12	times				0	
What is 12 times 7?	12	times	7			0	
What is 12 times 7?	12	times	7	is		0	
What is 12 times 7?	12	times	7	is	84	0	
What is 12 times 7?	12	times	7	is	84	end	1
The end-of-answer token: the episode ends						1	0 while the answer is still being written
						1	1 at the end: 84 is right. A wrong answer earns 0
						1	The sum of the rewards

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w}_{\text{Minus a small step}} - \underbrace{\alpha}_{\text{The direction that increases}} \underbrace{L(w)}_{\text{The loss}}$$

Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w}_{\text{Plus a small step}} + \underbrace{\alpha}_{\text{The direction that increases}} \underbrace{\mathbb{E}[r_1 + \dots + r_T]}_{\text{The expected sum of rewards}}$$

References

Ashish Vaswani and others. Attention is all you need. *Advances in Neural Information Processing Systems* 30, 2017.
arxiv.org/abs/1706.03762

Long Ouyang and others. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems* 35, 27730 to 27744, 2022.
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DeepSeek-AI. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* 645, 633 to 638, 2025.
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Zichen Liu and others. Understanding R1-Zero-like training: a critical perspective. *Conference on Language Modeling*, 2025.
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Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*, 2nd edition. MIT Press, 2018. Chapter 13.
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Daniel Jurafsky and James H. Martin. *Speech and Language Processing*, 3rd edition draft of 19 August 2026. Chapter 7, section 7.6, Decoding.

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Ari Holtzman and others. The curious case of neural text degeneration. *International Conference on Learning Representations*, 2020.

arxiv.org/abs/1904.09751

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Agentic AI, part 2

Agents

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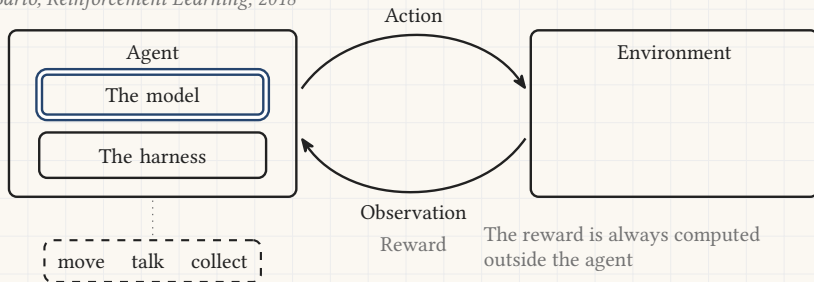
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Section 1

Agents and the loop

The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



The reward is always computed outside the agent

The spy game

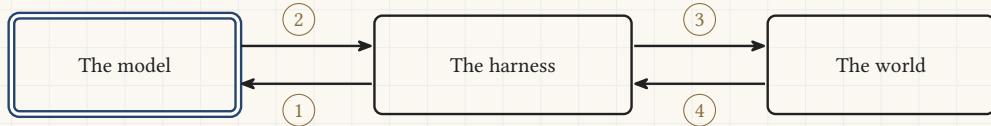
Environment	The 3 by 3 base
Observations	What the game prints after each move
Actions	Three tools, and one scan the harness runs
Reward	None is computed

The tax agent

Environment	Documents, a form, a mail account
Observations	What a tool returns
Actions	Read, extract, fill, send
Reward	None is computed

The harness

The program around the model. The model only writes text; the harness turns that text into actions and feeds back what happened.



In the spy game: `run_agent()`

- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call
- ③ Runs that tool on the world
- ④ Appends the result, then calls the model again or stops

*Anthropic, Claude API documentation,
tool use, 2026*

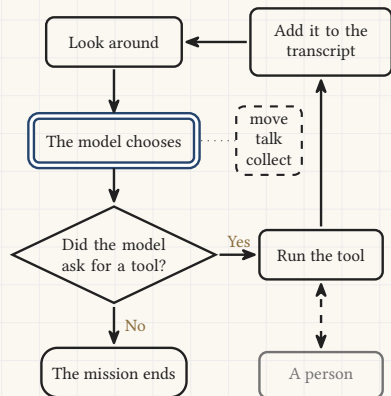
*Young, Effective harnesses for
long-running agents, Anthropic, 2025*

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Section 2

ReAct, played by hand

Thought, action, observation



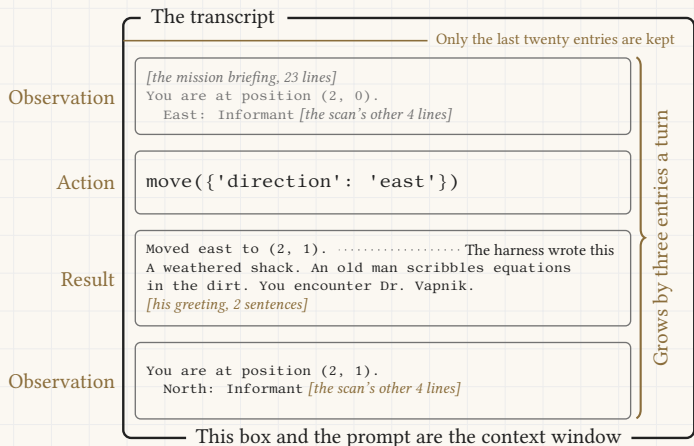
- The model writes a thought. Nothing outside changes.
- It writes one action. The harness runs it.
- What comes back is appended, and the model is called again.

```
while not mission_over(operative):  
    observation = look_around(operative, world)  
    action = decide(briefing, transcript, observation)  
    result = execute(action, tools)  
    remember(transcript, observation, action, result)
```

- A call to the model
- Something that is not a call to the model
- The tools the model may choose from
- The stopping test

The stopping test is in the loop's own condition. It is not a step counter. *Yao and others, ReAct, ICLR 2023*

The transcript the model is called with



Written once, by a person

You are a spy agent on a 3x3 grid.
Collect 3 dossiers in 30 turns.
Tools (use exactly one per turn):
TOOL: move(direction="north")
TOOL: talk(message="hello")
TOOL: collect()
Respond with exactly one TOOL: line.
Write one THOUGHT: line, then
one TOOL: line.

What the model wrote this step

TOOL: talk(message="Dr. Vapnik, I
seek the global minimum. Do you have
any jobs for an agent like me?")

gemini-2.5-flash, 16 September 2026
Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
context engineering, Anthropic, 2025

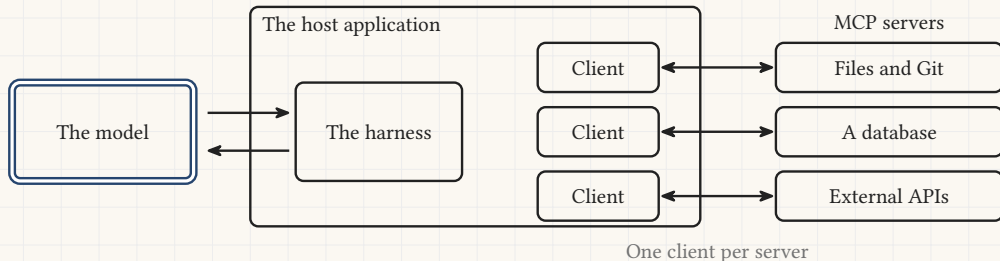
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Section 3

The Model Context Protocol

What MCP is

An open protocol: a standard plug between an application that uses a model and the servers that offer it tools and data.



Tools: functions the model may ask for, run by the server

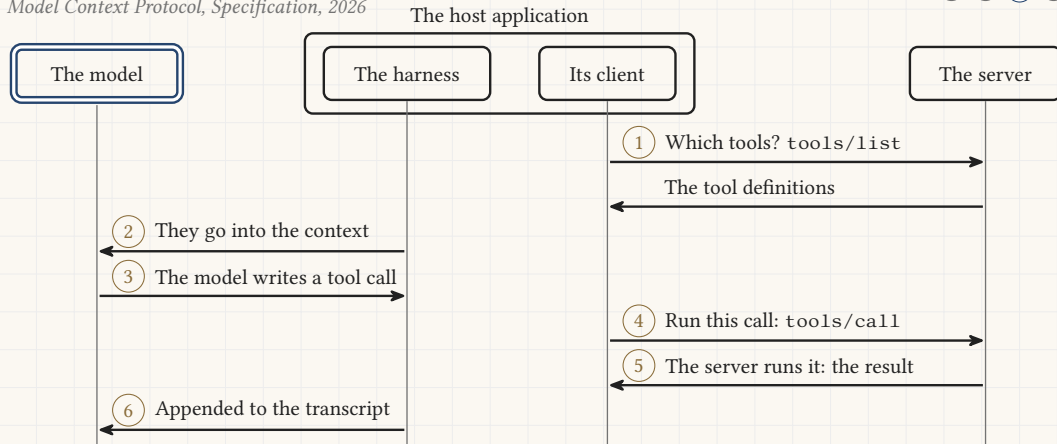
Resources: context and data, for the user or the model

Prompts: templated messages and workflows for the user

Model Context Protocol, Specification, 2026

How a tool call travels through MCP

Model Context Protocol, Specification, 2026

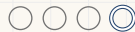


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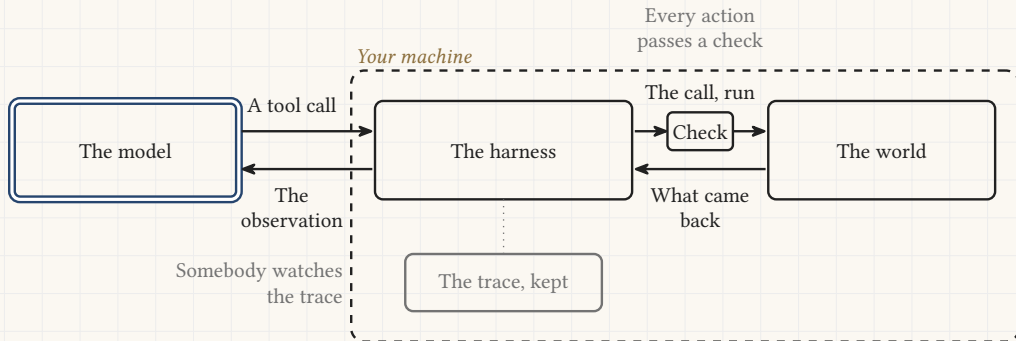
Section 4

The harness

An agent is a system



Cherny, Claude Code best practices, Anthropic, 2025
Anthropic, Claude API documentation, tool use, 2026



Play the spy game



[https://eth-bmai-hs26.github.io/
BMAI-PAGE/viz/we2/spy-game/](https://eth-bmai-hs26.github.io/BMAI-PAGE/viz/we2/spy-game/)

- 1 Open the page in your browser
- 2 Paste your OpenRouter key and press Save
- 3 Play with Manual, or choose Auto and press Run Agent

References: definitions and the loop

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*, 2nd edition. MIT Press, 2018.
incompleteideas.net/book/the-book-2nd.html

Erik Schluntz and Barry Zhang. Building effective agents. Anthropic, Engineering, 2024. anthropic.com/engineering/building-effective-agents

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Hugging Face. Agents Course, Unit 1, “Actions” and “Dummy agent library”. huggingface.co/learn/agents-course/unit1/actions

Prithvi Rajasekaran, Ethan Dixon, Carly Ryan and Jeremy Hadfield. Effective context engineering for AI agents. Anthropic, Engineering, 2025.
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Model Context Protocol. Specification. modelcontextprotocol.io/specification/latest

Justin Young. Effective harnesses for long-running agents. Anthropic, Engineering, 26 November 2025. anthropic.com/engineering/effective-harnesses-for-long-running-agents

Boris Cherny. Claude Code: best practices for agentic coding. Anthropic, Engineering, 18 April 2025. web.archive.org/web/20250420034405/anthropic.com/engineering/claude-code-best-practices